Industry Concentration Analysis Using Four-Firm Concentration Ratio (CR4) and Herfindahl – Hirschman Index (HHI)

Course: Managerial Economics

Instructor Name: Dr. Avisek Bhandari

By:-
Swastik Panda



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1. Introduction

Industry concentration refers to the extent to which total market output is dominated by a small number of firms. A highly concentrated industry indicates limited competition, strong market power, and entry barriers, whereas a low concentration level reflects competitive market conditions.

To measure concentration levels, two widely used indicators are:

- **Four-Firm Concentration Ratio (CR4)** – percentage share of market held by the top four firms.
- **Herfindahl-Hirschman Index (HHI)** – sum of squared market shares of all firms, indicating inequality of market distribution.

This report examines concentration trends in three major Indian sectors: the **Cement Industry**, the **Automobile Industry** and **Power Industry** between **2015 and 2024**.
The study aims to evaluate the dominance of leading firms, understand competitive intensity, and analyse structural changes over the decade.

2. Objectives of the Study

- Calculating the Four-Firm Concentration Ratio (CR4) for three industries.
- Computing the Herfindahl–Hirschman Index (HHI) for each year (2015–2024).
- Analysing market structure and competition using these values.
- Comparing concentration levels in the Cement, Automobile and Power sectors.
- Presenting insights through tables and graphs.

3. Methodology

3.1 Data Sources

Data was collected from trustworthy and publicly available financial and industry platforms:

- Company Annual Financial Reports
- Screener
- Moneycontrol
- Trendlyne
- SIAM (Society of Indian Automobile Manufacturers)

For all companies, **Revenue from Operations (Consolidated)** was considered.

3.2 Firms Selected

Cement Industry:

1. UltraTech Cement
2. Ambuja Cement
3. ACC Ltd
4. Shree Cement

Automobile Industry:

1. Maruti Suzuki
2. Tata Motors
3. Mahindra & Mahindra
4. Hyundai Motor India

Power Industry:

1. NTPC
2. Tata Power
3. Power Grid Corporation Of India Limited
4. Adani Power Limited

5.3 Formulas Used

Four-Firm Concentration Ratio (CR4):

$$CR4 = \frac{R_1 + R_2 + R_3 + R_4}{\text{Total Industry Sales}}$$

Where R1,R2,R3,R4 are revenues of top 4 companies and total industry sales were approximated using revenues of top 10 companies for each year.

Herfindahl–Hirschman Index (HHI):

$$HHI = \sum(\text{Market Share}^2)$$

where Market shares are in ratios (not percentage)

4. Data Presentation

Table 1: Cement Industry Data (2015-2024)

Year	UltraTech Share	Ambuja Share	ACC Share	Shree Share	CR4	HHI
2015	27.62985366	11.4247561	13.94239024	7.853512195	0.608505122	1150.001765
2016	25.4785	9.996347826	11.88740217	5.993065217	0.5335531522	926.3080933
2017	23.15268627	10.05059804	12.67740196	8.263794118	0.5414448039	866.0682164
2018	26.21658036	9.800892857	12.9263125	8.517535714	0.5746132143	1023.004556
2019	31.64021774	9.155451613	12.37347581	9.217790323	0.6238693548	1322.996235
2020	30.55973282	8.53051145	10.29529008	8.884854962	0.5827038931	1191.600541
2021	31.08393544	10.04658548	11.51847104	8.959277765	0.6160826972	1280.088755
2022	31.70671339	10.02865308	13.8785005	9.12121641	0.6473508338	1381.698921
2023	31.51633441	10.29175698	10.20144983	8.572418357	0.6058195959	1276.755532
2024	37.58536547	9.836941326	11.46123932	10.68966661	0.6957321273	1755.054091

Table 2: Automobile Industry Data (2015-2024)

Year	Maruti Share	Tata Share	Mahindra Share	Hyundai/ Hero Share	CR4	HHI
2015	12.96146667	9.5708	10.23776267	7.338133333	0.4010816267	418.259816
2016	14.47210256	10.85779231	10.35557692	7.271771795	0.4295724359	487.4500449
2017	16.12274699	10.44352289	10.45275422	6.787840964	0.4380686506	524.3449965
2018	17.75109091	13.05478409	10.93461818	7.179434091	0.4891992727	656.6387648
2019	17.47926316	14.47681684	11.14964211	6.973585263	0.5007930737	688.0482769
2020	14.9355	9.059533333	9.353735417	5.918422917	0.3926719167	427.6644005
2021	13.05139216	5.837072549	8.685676471	5.941868627	0.335160098	315.1570317
2022	13.99635	7.813495	9.389398333	4.78129	0.3598053333	367.9700526
2023	15.73437762	9.132704895	11.64353986	4.672197203	0.4118281958	488.378385
2024	16.06402381	8.660228571	11.5351869	4.4393	0.4069873929	485.8203413

Table 3: Power Industry Data (2015-2024)

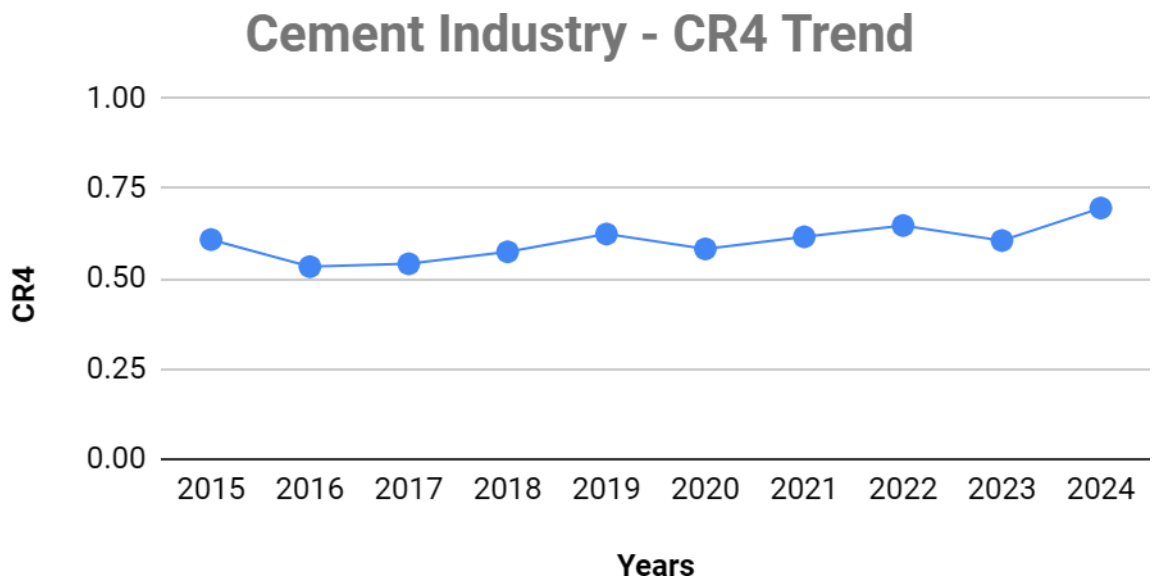
Year	NTPC Share	PGCIL Share	Tata Power Share	APL Share	CR4	HHI
2015	34.63045714	8.179633333	4.061090476	5.054357143	0.519255381	1308.213945
2016	31.24529333	9.215462222	3.762546667	5.63804	0.4986134222	1107.137352
2017	31.62254694	10.49390612	2.769257143	4.356089796	0.492418	1136.751844
2018	31.97219216	11.66763137	2.900603922	3.179482353	0.497199098	1176.877304
2019	33.00192279	12.45998897	2.984400735	0.8838970588	0.4933020956	1254.066155
2020	34.43227208	12.45304947	2.660551237	0.3551766784	0.4990104947	1347.864485
2021	34.15159655	12.8722931	2.086344828	0.1541965517	0.4926443103	1336.404088
2022	34.33487164	11.68382687	3.253143284	8.25519403	0.5752703582	1394.126391
2023	40.93406835	10.59685316	4.414544304	9.106589873	0.650520557	1890.30943
2024	37.72623765	9.666138824	4.66156	9.176298824	0.6123023529	1622.637848

Dataset Used:

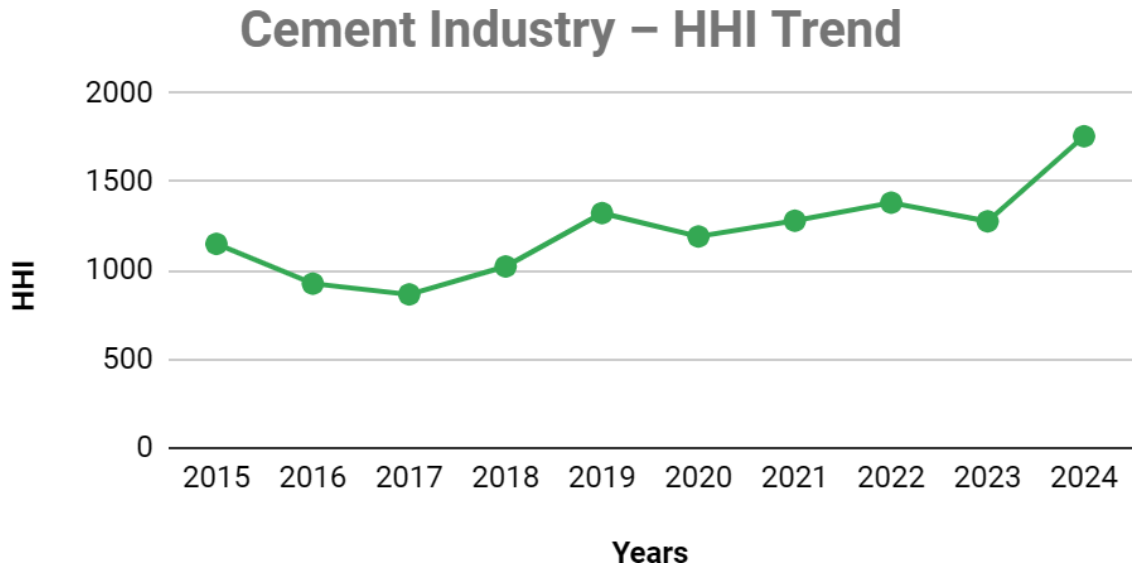
https://docs.google.com/spreadsheets/d/1YAcz0oDBVbiNeSoT8ckavvf8yeTBepQiPylxafho5JI/edit?gid=0#gid=0

5. Graphical Analysis

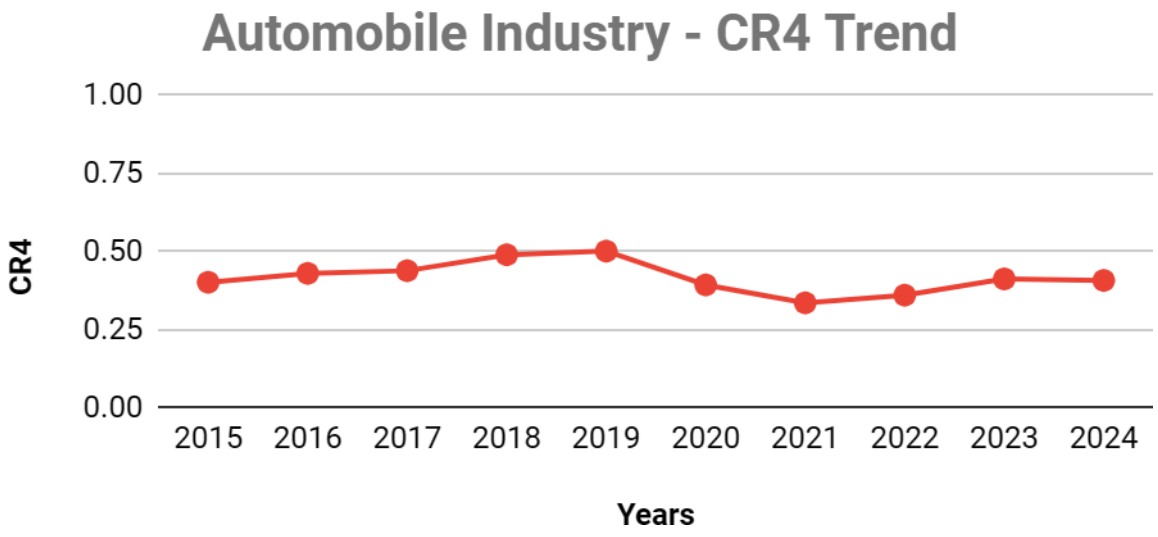
5.1 CR4 Vs Year (Cement Industry)



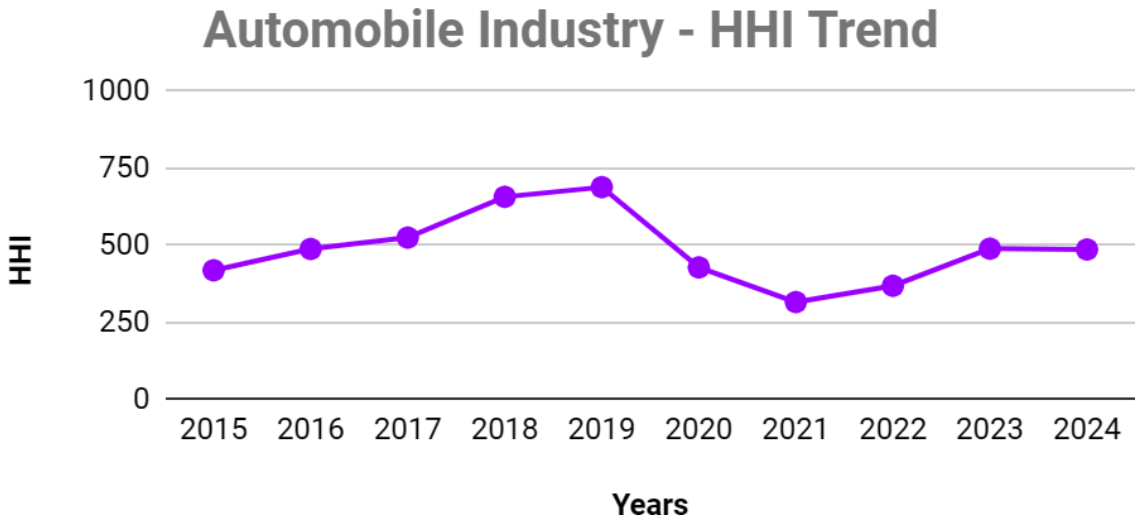
5.2 HHI Vs Year (Cement Industry)



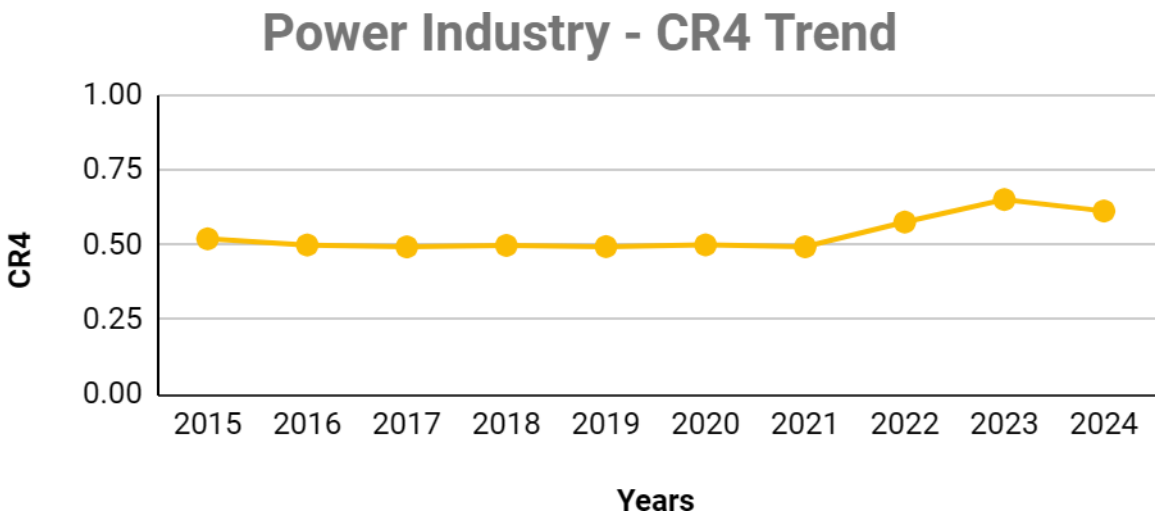
5.3 CR4 Vs Year (Automobile Industry)



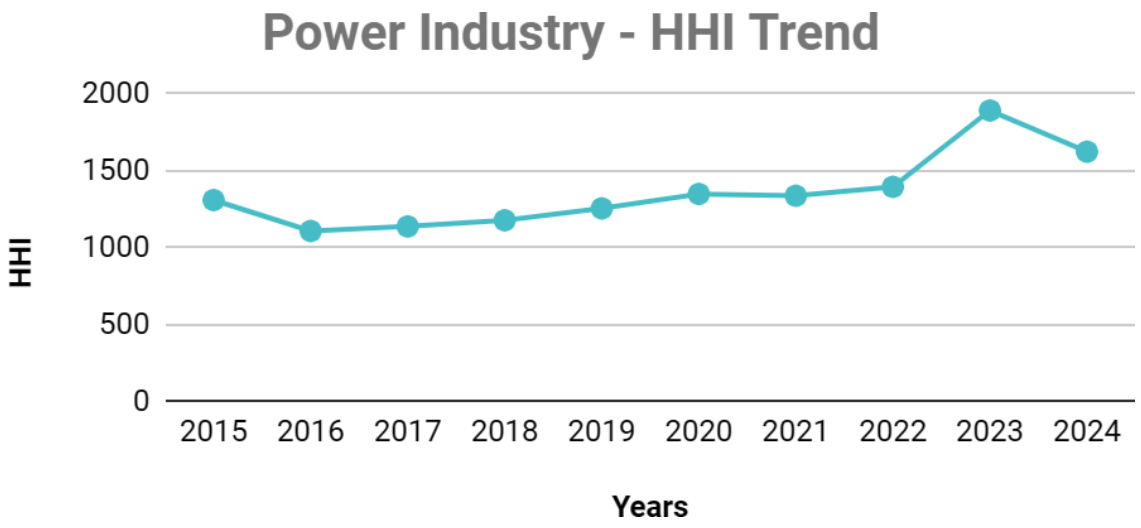
5.4 HHI Vs Year (Automobile Industry)



5.5 CR4 Vs Year (Power Industry)



5.5 HHI Vs Year (Power Industry)



6. Analysis & Interpretation

6.1 Cement Industry Analysis

Firms Analysed: UltraTech Cement, Ambuja Cements, ACC Ltd, Shree Cement

Timeframe: 2015 – 2024

1. Market Overview

For this part of the project, I have tracked the revenue of the top four players in the cement industry over the last ten years. The industry has definitely grown as the total sales more than doubled from about ₹82,000 Cr in 2015 to nearly ₹1.8 lakh Cr in 2024. The market power has also shifted apart from the sales growth. From the analysis, I found that the market is becoming less competitive and is more being dominated by the top firms.

2. Four-Firm Concentration Ratio (CR4) Trends

I calculated the CR4 to see how much of the total market is controlled by the four biggest companies.

- From my calculations I found that the ratio was around 61% in 2015 which actually dropped for a couple of years, hitting a low of 53% in 2016 This suggested that the smaller regional players were doing well at that time. However, from 2018 onwards, the trend has reversed. The big four players started capturing more market share every year. By 2024, the CR4 reached to 69.6%.
- This industry has now become concentrated since the ratio has crossed 60%. Seeing the ratio to be nearly 70%, I can say that the barriers to entry are getting higher every year. The top firms are dominating and squeezing out thereby making it really difficult for smaller players or new companies to enter the market.

3. Herfindahl-Hirschman Index (HHI) Analysis

While CR4 gives us a quick picture of the industry, I also looked at the HHI because it squares the market shares, which helps to highlight if one specific company is becoming too dominant.

- The HHI score stayed between 866 and 1,380 for most part of the decade. Technically, this is considered a "competitive" or "unconcentrated" market.
- However, there was a drastic change in the data of last year. In 2024, the HHI jumped up to 1,755.
- An HHI of over 1,500 puts the industry into the moderately concentrated category. The reason for this sudden spike is that the market leader, UltraTech, grew its share to almost 38%. Since HHI squares the market share, a single dominant firm like that pushes the index number up way faster than if the growth was shared equally among everyone.

4. Conclusion

Based on the data, the Indian Cement Industry isn't as competitive as it was ten years ago. It has now become a moderately concentrated industry. With the CR4 nearing 70% and the HHI finally breaking into the concentrated zone (above 1,500), the market power is heavily concentrated in the hands of the top players—specifically UltraTech. If this trend continues to grow, I can expect the top firms to have much more control over pricing in the future.

6.2 Automobile Industry Analysis

Firms Analysed: Maruti Suzuki, Tata Motors, Mahindra & Mahindra, Hyundai

Timeframe: 2015 – 2024

Market Overview

For the automobile sector, I tracked the revenue data of the top four manufacturers over the last decade. I found that the growth has been impressive in this sector. In 2015, the total industry sales were roughly 3.75 lakh crore rupees. By 2024, that number has gone up to 8.4 lakh crore rupees. This figure shows that the market size has more than doubled due to high demand. However, unlike other industries where the big players usually take over, the automobile industry has stayed very competitive with different firms.

CR4 Trends

- I have calculated the concentration ratio to see how much power the top four firms hold in the market. In 2015, the ratio was around 40%. It went up slightly to 50% in 2019 but then dropped significantly to around 33% in 2021. As of 2024, the ratio has settled back at around 41%.
- Since the top four firms control only about 40% of the market, it is clear that the industry isn't dominated by just a few giants. It means that while there are big players, they do not dominate the entire market. There is still a lot of market share, nearly 60%, left for other players which keeps the market open and active for newcomers.
- I also found that the automobile graph is much wavier that the other graphs. It shows that the share goes up and down (like the drop in 2021). This proves that market share in this sector is not

permanent, if a company stops innovating or a competitor launches a popular new model, the market share shifts immediately.

- I realized that tough competition is being provided by the global giants like Toyota, Honda, Kia, and MG which are part of the remaining 60%. This shows that international brands are entering India and successfully capturing market share, thereby proving that the barriers to entry are indeed low.
- Due to the low HHI, the top four firms cannot easily dictate prices. They have to constantly offer discounts, new features, or better mileage to keep customers.

HHI Analysis

- From the HHI scores, I can also see that this market is competitive and open for new players. Throughout the entire ten-year period, the HHI score remained very low. The highest it ever got was about 688 in 2019, and as of 2024, it is sitting at roughly 486.
- Since HHI score is below 1,500, I can consider it as an unconcentrated market. The automobile industry's score is consistently below 700, confirming that no single company has huge control in the market. Even though companies like Maruti and Tata are huge and have good demand, the calculation shows that their individual market power is balanced out by the competitors in the sector.

Conclusion

Based on the data, I can conclude that the Indian Automobile Industry is a highly competitive space. Unlike the cement industry which showed high concentration and demand of few players, the automobile sector has maintained a low concentration ratio over the last decade. With a CR4 of just 41% and a very low HHI, it is clear that

the market is not dominated by just few firms. This suggests that barriers to entry are lower compared to other sectors, and also there is healthy competition which generally benefits the consumers through better choices.

6.3 Power Industry Analysis

Firms Analysed: NTPC Limited, Power Grid Corp, Tata Power, Adani Power

Timeframe: 2015 – 2024

Market Overview

I have analysed the revenue data of the top four energy companies over the last ten years. There has been consistent growth in this industry. In 2015, the total industry sales were roughly 2.10 lakh crore rupees. By 2024, that number had doubled to 4.25 lakh crore rupees. This steady rise shows that the demand for electricity is constantly increasing as the country develops. However, a recent shift in the power industry shows that the top players are becoming more dominant.

CR4 Trends

- I calculated the concentration ratio to see how much control the top four firms have. For a long time, from 2015 to 2021, the ratio was very stable, staying around 49% to 51%. It was almost a flat line. However, recently, in 2022 and 2023, the ratio has jumped up significantly, peaking at roughly 65% in 2023 before settling at 61% in 2024.
- For a long time, the top four companies controlled only half of the market, which means that the industry was fairly balanced. But recently, their share crossed over 60%. This shows that the big players are taking larger share in the sales. I noticed this change happened largely because Adani Power bounced back.

After a huge drop in sales from 2019 to 2021, their revenue shot up in the last two years, which increased the total share of the top four firms.

- While NTPC has always been the leader, their recent revenue growth isn't just from burning more coal. They are also shifting to renewable energy like solar and wind. This Green Shift means the top players are looking towards the future, which will likely keep the market concentration high for the next decade as smaller coal plants shut down.

HHI Analysis

- The HHI scores also highlight the points that the CR4 show us. For eight years (2015–2021), the HHI was consistently low, staying between 1,100 and 1,350. This indicates that the market was quite competitive. But in 2023, there was a sharp spike where the score hit 1,890, and in 2024, it settled at 1,622.
- An HHI score between 1,500 and 2,500 shows that this industry is in the moderately concentrated category. The recent jump into this zone is mainly because of the market leader, NTPC. They consistently control a huge share (between 31% and 40%). When a single company owns such a large portion of the market, it automatically drives the HHI number up, showing that the industry is becoming less balanced.

Conclusion

I can conclude that the Indian Power Industry is currently in a transition phase. For most of the last decade, it was a fairly competitive market. However, the data from the last two years (2023-2024) shows that it is moving towards oligopoly. With the CR4 crossing the 60% mark and the HHI entering the concentrated zone, the market power is getting concentrated in the hands of few players. This is mainly due to the dominance of the state-run giant NTPC and the rapid recovery of

private players like Adani Power. If this trend continues, new players might find it harder to enter this industry in the future.

6.4 Cross-Industry Comparison

Looking at the period from 2015 to 2024, the three industries differ quite noticeably in how concentrated their markets are.

The cement industry comes across as the most consolidated overall. Its CR4 stays close to 0.60, moving between roughly 0.53 and 0.70, while its HHI sits in the 866–1,755 range, averaging around 1,217.

The power sector shows a slightly lower average CR4 of about 0.53 (anywhere from 0.49 to 0.65), yet its HHI turns out to be higher, with an average near 1,357 and values rising steadily from around 1,107 to almost 1,890. This indicates that the market share among the leading companies is more uneven than in cement—meaning one or two firms probably command a much larger share.

On the other hand, the automobile industry stands at the opposite end of the spectrum. It is clearly the least concentrated of the three, with CR4 values shifting between 0.34 and 0.50 (average close to 0.42) and consistently low HHI figures, mostly between 315 and 688 (average roughly 486). This pattern fits well with a competitive market structure where several firms actively compete and no single company dominates.

Using the common benchmarks in industrial-organisation literature, automobiles fall into the unconcentrated / strongly competitive category. Both cement and power fit the moderately concentrated, oligopolistic range. However, because the power industry has a higher HHI despite a CR4 similar to cement, its leading firms are less evenly matched.

6.5. Discussion of Structural Changes

When you look at the cement industry across these years, the overall trend is pretty clear. There's a small dip around 2016–17, but after that the numbers just keep climbing. CR4 slowly goes from something like 0.53 to almost 0.70, and the HHI also climbs from the mid-800s to well over 1,700 by 2024. This basically means the big companies end up controlling more of the market as time passes—maybe because they expanded, bought out others, or simply because the smaller players didn't keep up.

The automobile sector behaves very differently. At first (2015–2019), it looks like concentration is rising—CR4 moves from around 0.40 to 0.50, and HHI also goes up. But right after that, especially around 2020, everything drops sharply. HHI even touches around 315 in 2021, which is extremely low. The last couple of years show a slight recovery, but overall, the long-term picture is that the auto market becomes more open and competitive. New brands, more variety, EVs, compact SUVs—all of this spreads the market across more firms.

The power sector is fairly calm for most of the early period. Up to around 2021, CR4 is hovering around 0.49–0.52, and HHI stays in a narrow band around 1,100–1,350. Nothing dramatic there—it's a steady oligopoly. But after 2022, things change quickly. CR4 suddenly goes above 0.57, reaches around 0.65 at one point, and HHI shoots up close to 1,890 in 2023. It settles a bit afterwards, but the jump clearly suggests that a few major players became a lot stronger than the rest.

So, putting everything together:

cement and power move toward greater concentration over time, while the automobile market becomes more spread out and competitive.

7. Conclusion

A comparison of the CR4 and HHI values for the three industries clearly shows how different their market structures are within India. The cement and power sectors are gradually becoming more concentrated, with a handful of large companies starting to dominate. This points towards these industries slowly moving into a tighter oligopoly. In contrast, the automobile sector is still quite competitive. The industry is moreover quick paced reflecting the entry of new markets due to its decreasing concentration levels.

From a policy point of view, sectors like cement and power may need more careful oversight. When these industries start getting dominated by a few big players, it can lead to problems like higher prices or making it tougher for smaller companies to compete. But at the same time, having a few strong firms can also make things run more smoothly and keep the sector stable, which really matters for areas like infrastructure and energy. For the automobile industry, the focus should be on keeping competition healthy and encouraging innovation, particularly in areas like electric vehicles and green mobility, which are becoming increasingly important.

In short, the analysis shows that each industry reacts differently to changes in regulation, technology, and market behaviour. This highlights the need for competition policies that are tailored to the specific needs and future direction of each sector, rather than applying a single uniform approach

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